

SUSAN WAITHIRA NYAWIRA

Contact: +254 758 796 429 | Email: sw.nyawira@gmail.com | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Computer Science graduate and Software Engineering Trainee with practical experience in software development, backend systems, RESTful APIs, databases and ICT operations. Hands-on experience developing web and data-driven applications while applying software engineering, testing, version control and DevOps practices. Interested in using technology to build secure, reliable and scalable digital solutions that address real-world problems.

TECHNICAL SKILLS

Languages: JavaScript, Java, Elixir

Frameworks/Libraries: React, Node.js, Express, Spring Boot, Phoenix, Tailwind CSS

Security Concepts: User Authentication, Input Validation, Secure System Design, JWT

Tools & Platforms: Git, GitHub, Docker, Postman, VS Code, Microsoft Office Suite

Core Concepts: Data Structures, Object-Oriented Programming (OOP), SDLC, Networking Fundamentals, CRUD, API Integration, Agile, SSADM

Databases: MySQL / PostgreSQL

PROJECTS

Personal Portfolio Website

React + Tailwind CSS

- A responsive web application with structured content to enhance accessibility and optimise page layout for cross-device performance.
- [Portfolio](#)

Hesabu — Financial Systems Tool

Node/Express backend

- Designed and implemented a modular financial system for calculating transaction fees, structuring the architecture into a logic engine, UI layer and API-ready backend.
- Focused on accessibility and usability, ensuring the tool caters to non-technical users.
- Designed with scalability and future API integration in mind.
- [HESABU](#)

Authentication System

Node.js, JWT

- Built a secure authentication system using JWT-based user sessions.
- Implemented password hashing and input validation to enhance data security.
- Tested endpoints and ensured proper handling of user authentication and session management.

WORK EXPERIENCE

Software Engineering Trainee — Tracom Academy 2026 – Present

Tracom Services Limited

- Currently undergoing practical training in software engineering, with exposure to backend development, web development, mobile development, databases, APIs, DevOps and software development practices.
- Develop and test applications using modern programming languages, frameworks and development tools.
- Build RESTful APIs and work with relational databases as part of backend development projects.
- Apply Git and collaborative software development practices to manage and maintain source code.

- Gain practical exposure to containerization and application deployment using Docker.

ICT Support & Operations Attachment 2025

CIC Insurance Group — BuruBuru Branch

- Supported day-to-day ICT operations and provided technical assistance to staff, handling policy processing systems and client service workflows.
- Assisted with troubleshooting computer hardware, software and connectivity-related issues, maintaining continuity of daily operations.
- Collaborated with cross-functional teams (customer care, finance, CX) to ensure accurate data handling.
- Gained practical exposure to ICT operations and real-world system usage within a corporate environment.

Coding Reviewer 2023 – 2024

African Centre for Women in ICT (ACWICT)

- Reviewed codebases and provided structured feedback to improve readability and maintainability.
- Collaborated in iterative development workflows, contributing to continuous improvement.
- Identified recurring issues and documented best practice recommendations.

Byte Scan Operator 2022 – 2023

Agricultural Society of Kenya

- Managed digital ticketing systems during high-volume public events.
- Diagnosed and resolved real-time operational issues under pressure.
- Maintained system accuracy and efficiency across multiple event cycles.

EDUCATION

Bachelor of Science in Computer Science — Expected Graduation Year: 2026

Technical University of Mombasa

Relevant Coursework: Data Structures & Algorithms, Computer Networks, Database Systems, Software Engineering, Web Development

LEADERSHIP & ACTIVITIES

Student Representative

- Represented student interests and coordinated academic engagement initiatives.

Team Manager — Group Projects

- Led student teams in delivering software projects, managing task allocation and timelines.

REFERENCES

Available upon request